

LESSON 2-4 • HOMEWORK PACKET • SHEET 1 OF 2**Building a Box Plot**

Maryland Standard 6.SP.B.4 • Learning target – I can find the five-number summary of a data set and use it to build a box plot.

REMEMBER

- **Five-number summary:** minimum, Q1, median, Q3, maximum.
- **Q1** is the median of the lower half; **Q3** of the upper half. Odd count: leave the median out.
- The **box** goes from Q1 to Q3, with a line at the median. **Whiskers** reach the min and max.

WORKED EXAMPLE

Find the five-number summary of 4, 6, 9, 10, 13, 15, 18.

Step 1 Min = 4, max = 18, median = 10

Step 2 Lower half 4, 6, 9, so Q1 = 6

Step 3 Upper half 13, 15, 18, so Q3 = 15

Answer 4, 6, 10, 15, 18

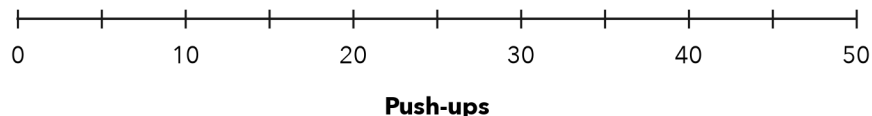
PRACTICE

show your work • label units

1. **Push-ups.** Nine students did 22, 40, 10, 34, 18, 46, 25, 14, and 30 push-ups. Find the five-number summary.

Min	Q1	Median	Q3	Max

2. Draw a box plot of the push-up data from Problem 1.

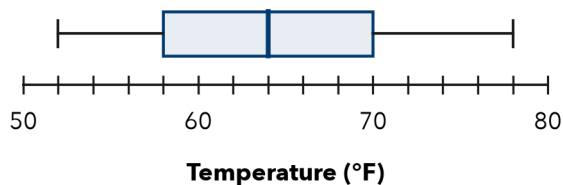


3. **Temperatures.** The box plot shows daily high temperatures (°F) for a month.

a. Median: _____ °F

b. Q1 and Q3: _____

c. Fraction of days from 58 to 70°F: _____



4. **Homework Minutes.** 26, 14, 39, 20, 11, 32, 18, 28. Find each value.

a. Q1: _____

b. Median: _____

c. Q3: _____

5. **Find the Error.** Grace says Q1 of 2, 5, 7, 8, 12, 14, 20 is 6. She used 2, 5, 7, 8 as the lower half. Explain her error and find Q1.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Explain to someone at home what the box in a box plot shows.

LESSON 2-4 • HOMEWORK PACKET • SHEET 2 OF 2

Comparing Box Plots

Maryland Standard 6.SP.B.4 • Learning target – I can compare two groups by reading the medians and box widths of their box plots.

REMEMBER

- Compare **centers**: use the medians.
- Compare **spread**: box width and range.
- A **narrower box** ($Q3 - Q1$) is more **consistent**.

WORKED EXAMPLE

A: median 18, box 14–22. B: median 24, box 20–26.

Step 1 Center: $24 > 18$, so B scored higher.

Step 2 Box widths: A is $22 - 14 = 8$; B is $26 - 20 = 6$.

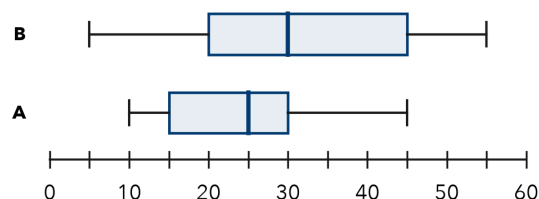
Answer B is higher and more consistent

PRACTICE

show your work • label units

1. **Reading Time.** Minutes of reading per day for Class A and Class B.

- a. Greater median: _____ by _____ min
- b. Box widths: A _____ B _____
- c. More consistent: _____



2. Find each measure of spread.

- a. $Q1 = 12$, $Q3 = 30$. Box width: _____
- b. $Min = 5$, $max = 41$. Range: _____

3. **Bowling.** Complete the table for two bowlers' scores.

	Min	Q1	Median	Q3	Max	Box width	Range
Maya	80	95	102	110	125		
Leon	70	90	105	112	140		

Who is more consistent? _____

4. **Plant Heights.** Both medians are 14 cm. Box S: 12–16 cm. Box T: 8–20 cm. Which varies more?

☐ Group S

☐ Group T

5. **Choose a Bus.** Both medians are 20 min. Boxes: A 18–22, B 14–26 min. Which is more reliable?

TEST PRACTICE

choose the best answer

6. X: $Q1 = 22$, $Q3 = 34$. Y: $Q1 = 25$, $Q3 = 31$. Which middle half is more consistent?

A X

B Y

C They are equal

D Cannot tell

7. A box plot has the summary 8, 12, 15, 21, 30. How wide is the box?

A 9

B 22

C 15

D 3

CHECK YOURSELF

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☐ I need a little help

☐ I have a question about #_____